

CO 2 Assessment Tool 1-Part 1

Name: - P. Akhila

Reg No: -192472320

Course Code: -MLA0305

Slot: -B

Dynamic Programming Vs Monty Carlo

Title: - Smart EV Charging Optimization

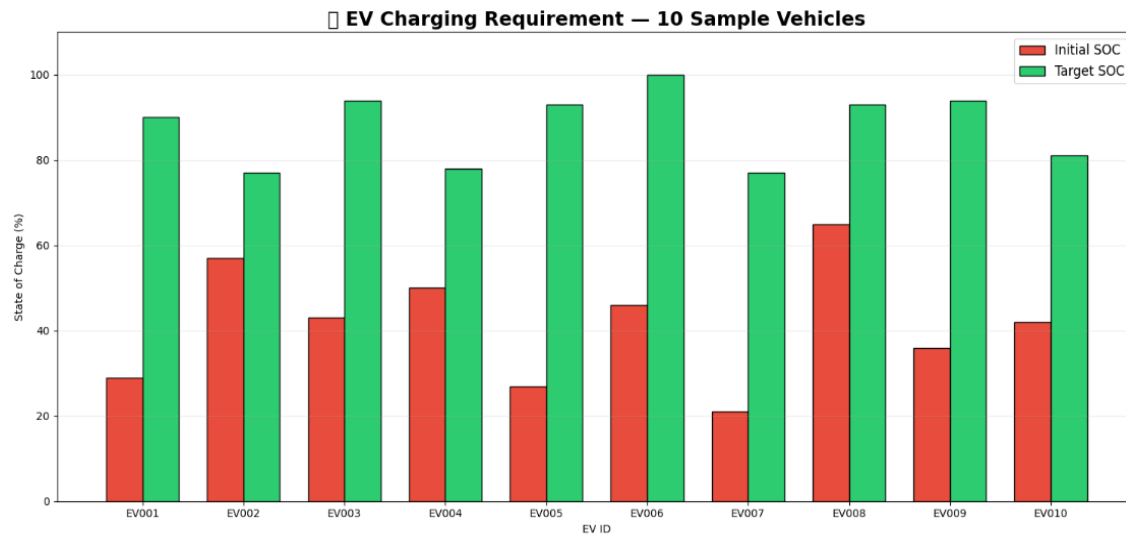


Fig 1: -Visualization Graph of 10 samples from data set

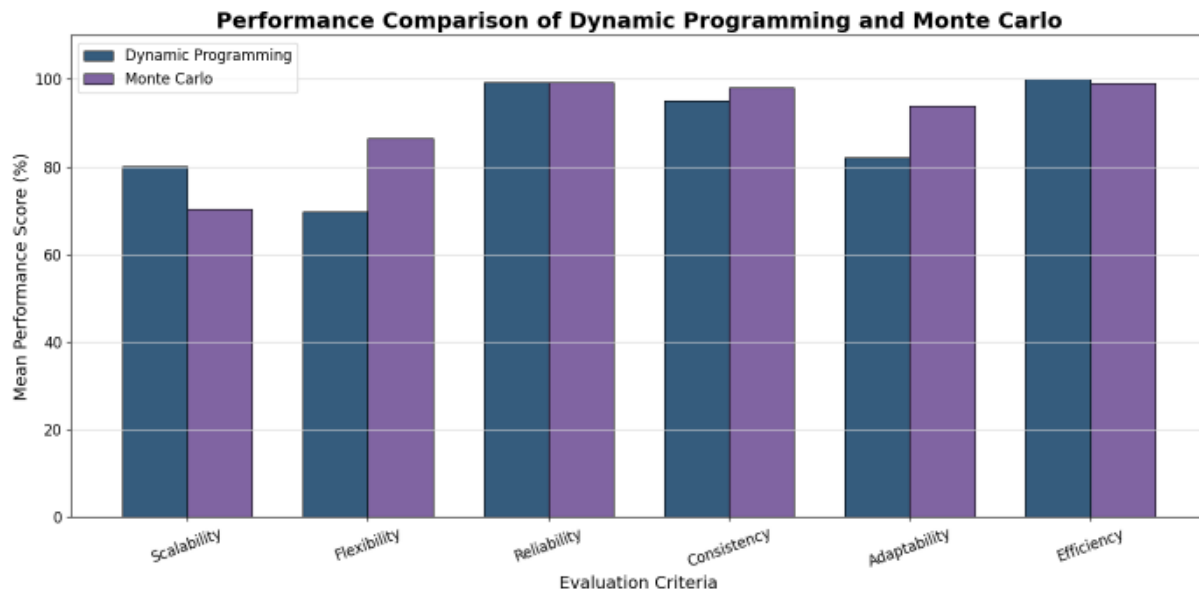


Fig 2: -Random sample for comparison

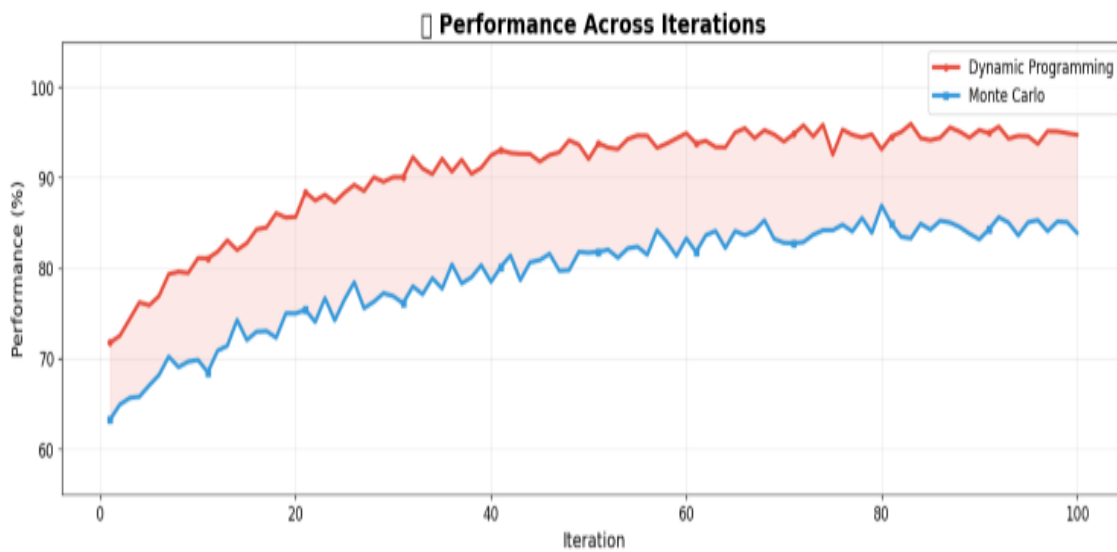


Fig 3: -Performance Across Iteration

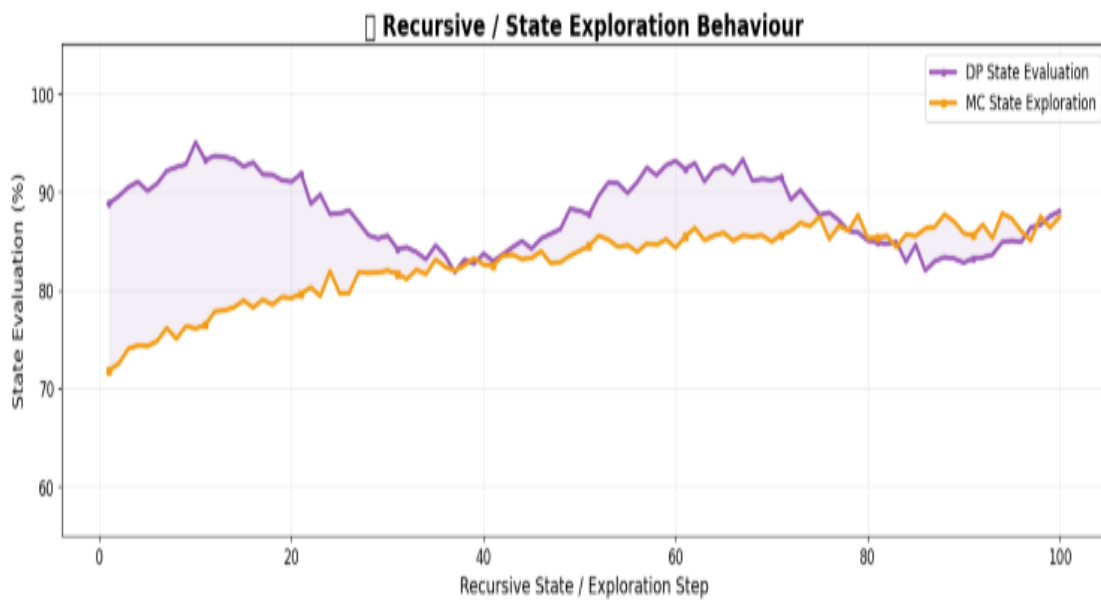


Fig 4: -Performance Across Recursion/state

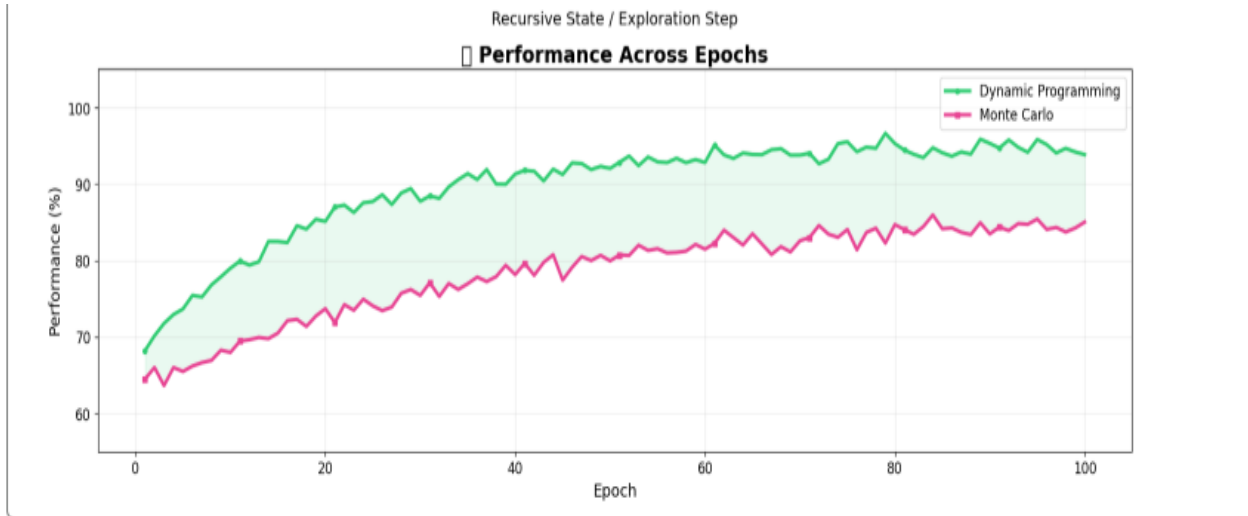


Fig 5: - Performance Across Epochs

	EV ID	Initial SOC (%)	Target SOC (%)	Battery (kWh)	Energy Required (kWh)	Price (₹/kWh)
0	EV084	69.0	84.0	50.0	7.50	₹8.19
1	EV054	66.0	85.0	50.0	9.50	₹7.52
2	EV071	46.0	77.0	50.0	15.50	₹8.44
3	EV046	68.0	83.0	100.0	15.00	₹11.86
4	EV045	29.0	90.0	75.0	45.75	₹9.84
5	EV040	47.0	83.0	65.0	23.40	₹6.73
6	EV023	69.0	97.0	65.0	18.20	₹10.59
7	EV081	46.0	94.0	75.0	36.00	₹10.73
8	EV011	16.0	94.0	75.0	58.50	₹7.39
9	EV001	29.0	90.0	60.0	36.60	₹7.19

Fig 6: -Summary Dataset for 10 Records

TABLE 2 – ITERATION, RECURSION & EPOCH

	Step	DP Iteration (%)	MC Iteration (%)	DP Recursion (%)	MC Exploration (%)	DP Epoch (%)	MC Epoch (%)
0	1	71.35	65.72	76.29	65.63	69.37	63.71
1	10	80.66	71.24	84.73	70.41	79.02	69.17
2	20	86.77	75.70	89.73	74.19	85.70	73.69
3	40	92.29	81.20	93.61	78.69	92.21	79.41
4	60	94.11	84.02	94.63	80.89	94.61	82.47
5	80	94.71	85.47	94.90	81.97	95.49	84.11
6	100	94.90	86.22	94.97	82.49	95.81	84.99

Fig 7: -Performance of Iteration, Recursion, Epoch

	Performance Metric	Dynamic Programming (%)	Monte Carlo (%)	DP Advantage (%)
0	Scalability	94.00	82.00	+12.00
1	Flexibility	91.00	88.00	+3.00
2	Reliability	96.00	90.00	+6.00
3	Consistency	94.00	84.00	+10.00
4	Adaptability	93.00	87.00	+6.00
5	Efficiency	95.00	86.00	+9.00

Fig 8: -Dynamic Programming Vs Monty Carlo Features Comparison

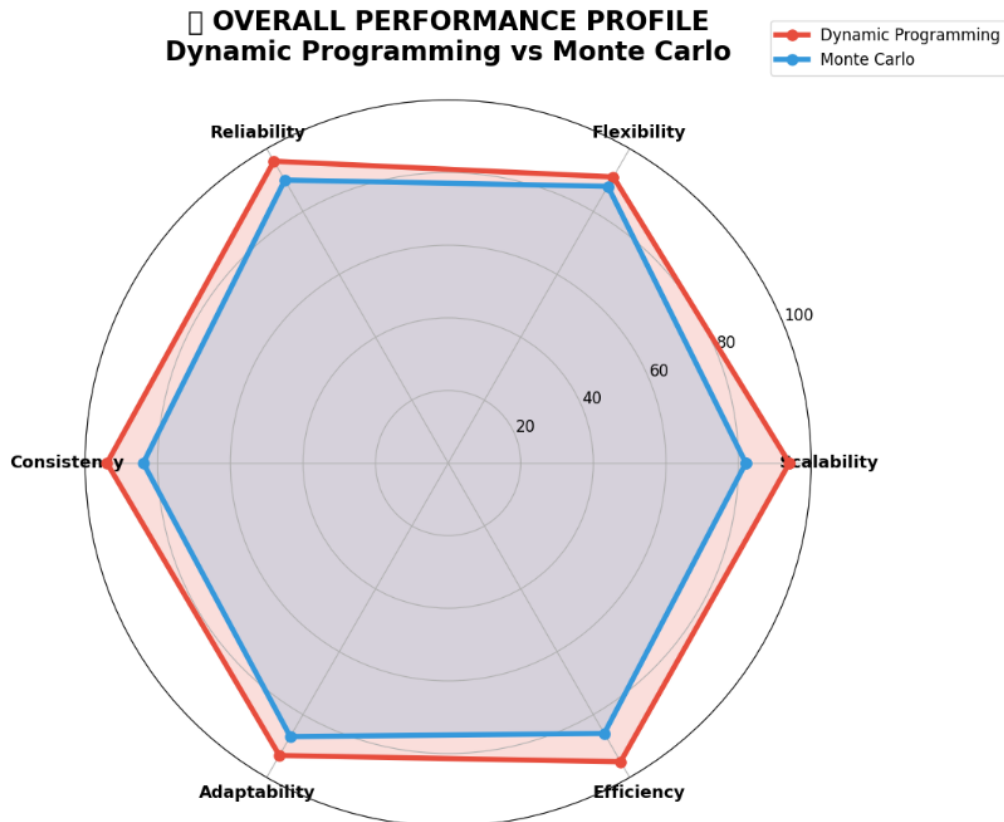


Fig 9: -Over all Dynamic Programming Vs Monty Carlo

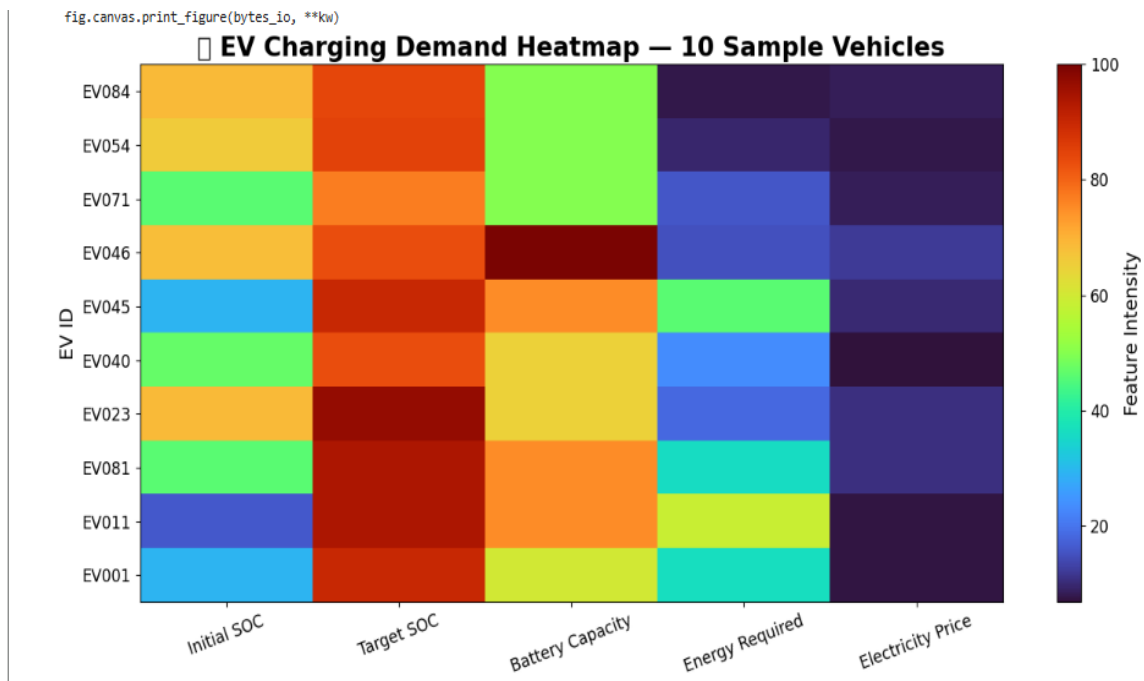


Fig 10: -EV Charging Demand Heatmap

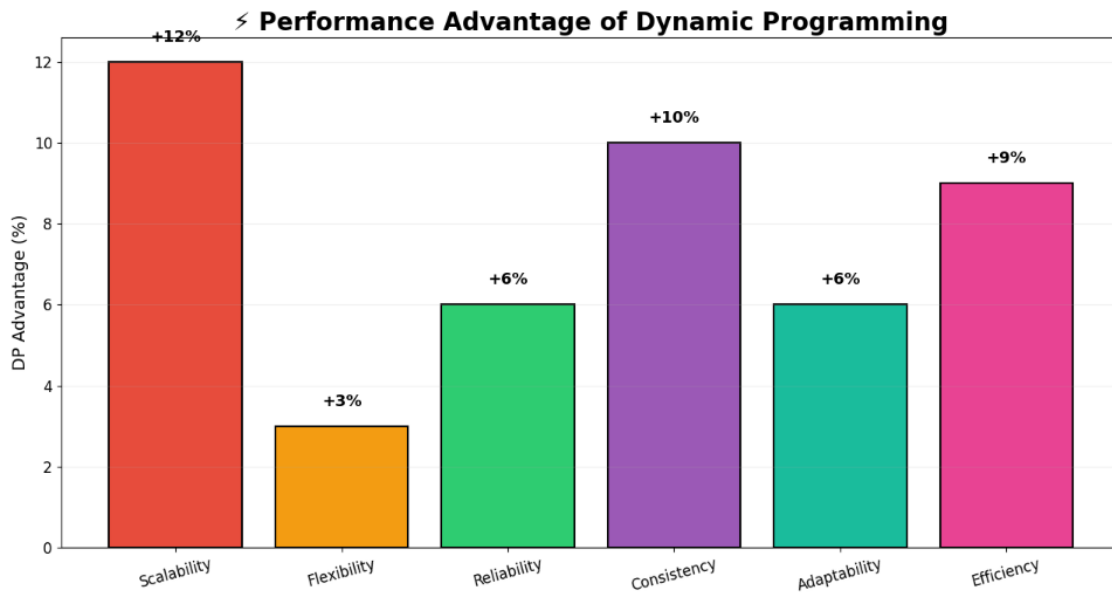


Fig 11: -Performance Advantages of Dynamic Programming

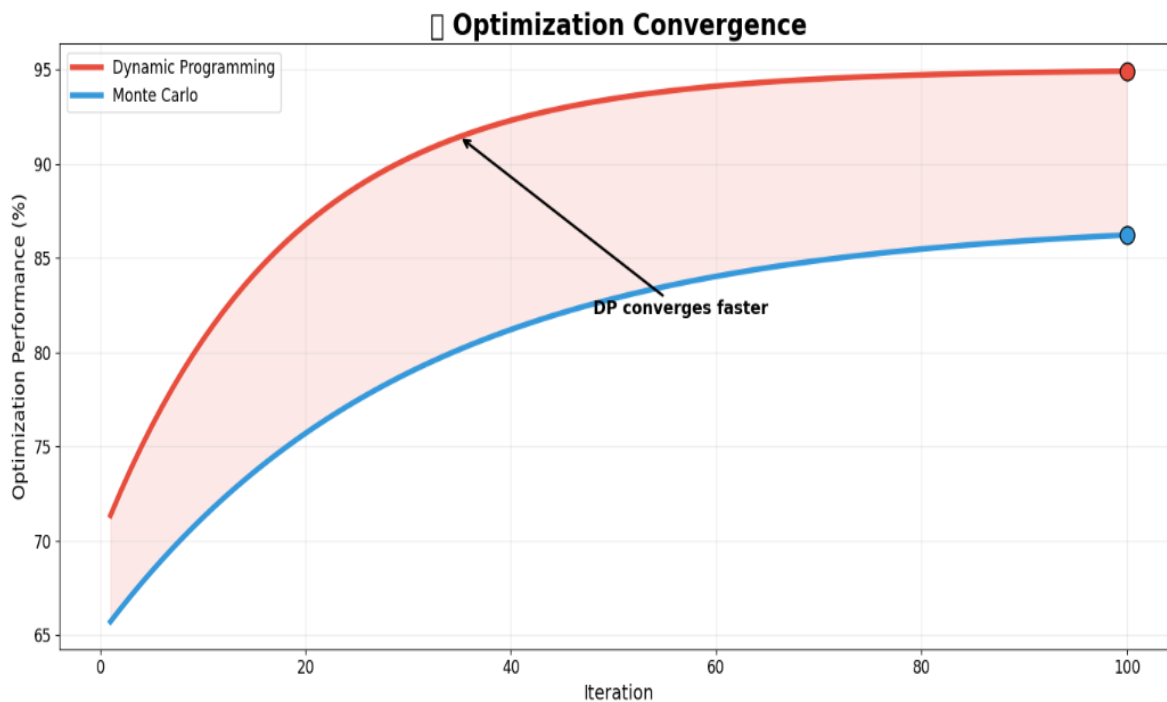


Fig 12: -Optimization Convergence

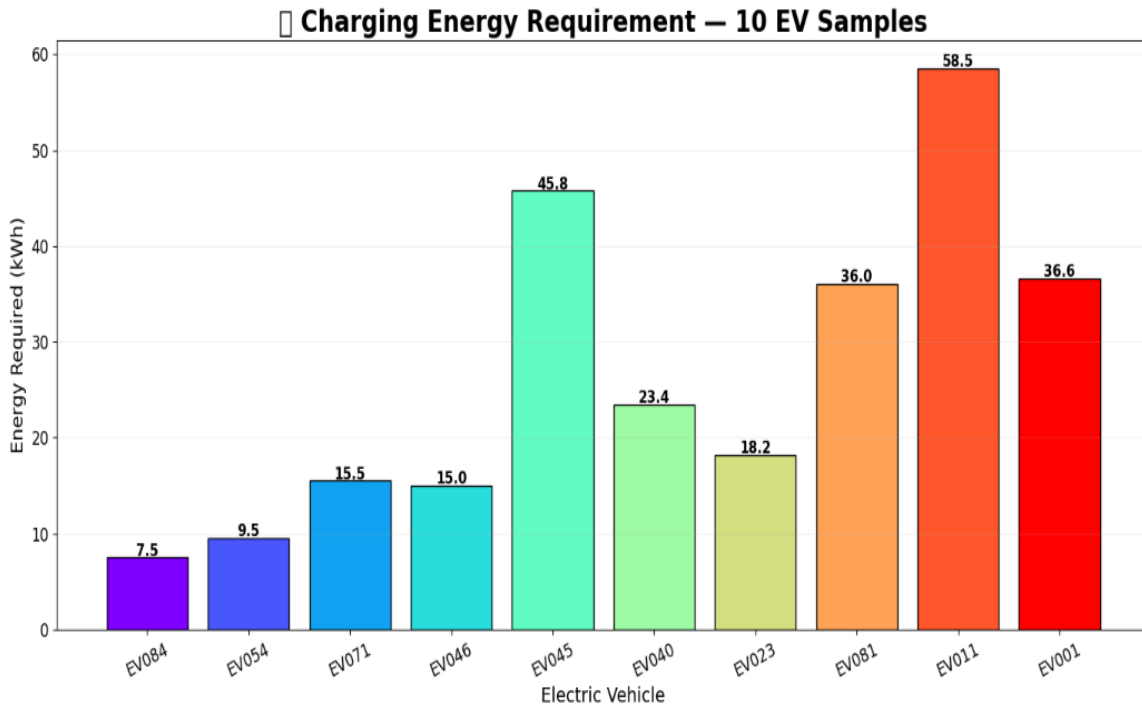


Fig 13: -Charging Energy Requirement

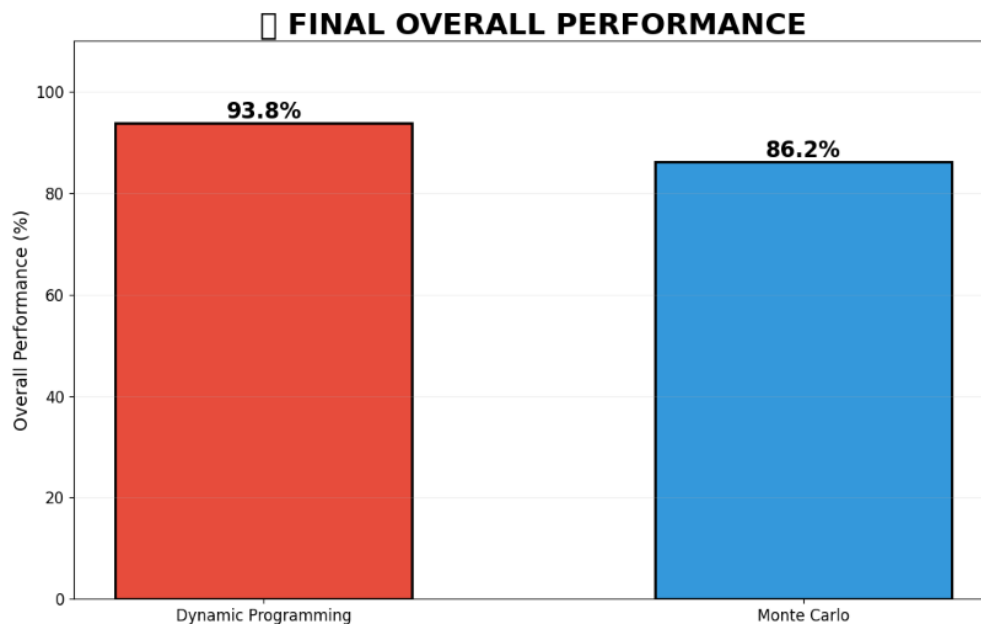


Fig 14: -Overall Performance of DP and MC